

Chapter 13

Direct Price Discrimination

Learning Objectives

- 13.1 Describe the motivation for price discrimination.
- 13.2 Discuss the practice of direct price discrimination.
- 13.3 Explain how arbitrage can be prevented using direct price discrimination.
- 13.4 Describe how the Robinson-Patman Act prohibits price discrimination against small retailers.
- 13.5 Interpret how price discrimination is implemented using real-life business cases.

Chapter 13 – Main Points

- If a seller can identify two groups of consumers with different demand elasticities, and it can prevent arbitrage between two groups, it can increase profit by charging a higher price to group with the less-elastic demand.
- **Price discrimination** is the practice of charging different people or groups of people different prices that are not cost-justified. Typically, more people are served under price discrimination than under a uniform price.
- **Arbitrage** can defeat a price discrimination scheme if enough of those who purchase at low prices re-sell to high-value consumers. This can force a seller to go back to a uniform price.
- A **direct price discrimination scheme** is one in which we can identify members of the low-value (more price elastic) group, charge them a lower price, and prevent them from re-selling their lower-priced goods to the higher-value group.
- It can be illegal for business to price discriminate when selling goods (not services) to other businesses unless
 - price discounts are cost-justified, or
 - discounts are offered to meet competitors' prices.
- Price discrimination schemes may annoy customers who know they're paying more than others and can make them less willing to buy because they know someone else is getting a better price. If you can, keep them secret.

Warm-Up

- Where have you noticed the price of a good be different based on some characteristic of the buyer?

Price Discrimination

- This chapter moves beyond pricing products for all consumer to equally consume and instead introduces a method that tries to extract the willingness to pay of different consumers
- **Price discrimination** is the practice of selling the same good at different prices to two or more groups of consumers based on differences in demand
- Firms will and do spend money determining what their consumer base is in order to determine better pricing decisions.
- By price discriminating, firms turn consumer surplus into producer surplus

Price Discrimination

- The key to price discrimination is to have two (or more) groups of consumers with different elasticities
- The goal of price discrimination is to turn consumer surplus into producer surplus
- Coupons and sales are good ways to determine low value consumers
- Need to be able to determine demand functions
- Trial and error can be used
- If marginal revenue is too high in one market, move/sell more in the other market
- Are all price differences price discrimination?
 - No. Some price differences reflect the cost involved in a sale.
 - E.g., Selling a one-year magazine subscription that mails to the house vs. paying for the same magazine at a newsstand or bookstore.

Discussion Activity 1

- Why would a firm want to charge different prices to different consumers for the same product?
- Think about different willingness to pay and how to convert consumer surplus into profit.

Types of Price Discrimination

- **First degree (or perfect price) discrimination - direct**

- The practice of charging each consumer a different price and is direct.
 - Extracts all consumer surplus from the market, and turns it into producer surplus
 - Efficient, but it hurts consumers
 - Universities almost do this

- **Second degree price (quantity) discrimination - indirect**

- Charging consumers different prices for different quantities of the same good
 - E.g. city water, bulk goods
 - It is indirect price discrimination

- **Third degree (multimarket) price discrimination - direct**

- The practice of charging groups of consumers with different elasticities different prices.
 - E.g. Movie tickets, early bird buffets
 - It is direct price discrimination

Implementing Price Discrimination

If a firm is attempting to directly price discriminate, it must be able to identify the differing demands of the consumer groups.

The firm must have these condition hold:

- **The firm must have market power**
 - So that the firm doesn't have to take the market price as a given
- **The elasticity of demand must be different**
 - You charge the more inelastic group a higher price
 - Due them being less responsive to changes in price
 - You charge the more elastic group a lower price
 - Due to them being more responsive to changes in price
- **A firm must be able to prevent/limit arbitrage**
 - This prevents lower value consumers from selling to higher value consumers.
 - Devise schemes to alter timing of consumption
 - Make the transaction cost of resale higher

Implementing Price Discrimination

- Student Discounts
 - Require student IDs or email
- Senior Citizen Pricing
 - Offer discounts based on individual's age
- Geographic Pricing
 - Offer lower pricing to customers in different regions
- Resident v. Tourist Pricing
 - Local residents get lower prices than tourists (e.g. national and theme parks)
- Business v. Personal Customers
 - Businesses may pay more for licenses to software
- Online v. In-store
 - Stores may charge more for online purchases (e.g. Doordash) than in-store purchases

Activity – Design a Direct Price Discrimination Strategy

Scenario:

- You're helping launch a subscription-based educational streaming platform. You're exploring a segmented pricing model with these options:
 - Student Plan
 - Family Plan
 - Professional/Enterprise Plan
 - Regional Pricing for Emerging Markets

Instructions:

1. Identify demand elasticity for each segment. Which groups are likely to be more price-sensitive?
2. Propose pricing tiers. Set rough price points and justify them based on willingness to pay.
3. Prevent arbitrage. Suggest policies or tools to prevent customers from misusing lower-tier pricing.
4. Evaluate fairness or risk. Are there risks related to brand perception, legal issues, or customer backlash?

Activity – Design a Direct Price Discrimination Strategy

1. Elasticity by Segment:

1. Students – highly elastic, low budgets and many alternatives
2. Families – moderately elastic, share accounts
3. Professionals – less elastic, higher WTP and wants features
4. Emerging markets – highly elastic, lower income and sensitivity to exchange rates

2. Sample Tier Pricing:

1. Student - \$4.99/month
2. Family - \$9.99/month (up to 5 users)
3. Pro/Enterprise - \$24.99 - \$49.99/month depending on features/users
4. Emerging Market - \$2.99/month

Activity – Design a Direct Price Discrimination Strategy

3. Preventing Arbitrage

- Students – ID verification
- Family – terms of use for limiting sharing
- Professionals – API access restrictions
- Emerging Markets – Geolocation/IP tracking

4. Risks

- Fairness concerns since some users pay less
- Regulatory issues (Robinson-Patman Act discussed later)
- Technical workarounds (VPN, account sharing)

Discussion Activity 2

- If you offer student discounts, how could students exploit this system, and how would you stop them?
- Think about resale, account sharing, or eligibility proof (IDs, email, SheerID, address).

Case Study: Spotify's Direct Price Discrimination Strategy

Spotify serves a global audience with vastly different:

- Incomes
- Use cases (personal vs. shared)
- Price sensitivities
- To maximize revenue and adoption, Spotify uses direct price discrimination across demographic and usage segments.

Plan Type	Target Segment	Monthly Price (US)	Eligibility/Verification	Why It Works
Student	College students	\$5.99	.edu email, SheerID	High elasticity, low income
Individual	General consumer	\$10.99	None	Baseline willingness to pay
Duo	Couples/roommates	\$14.99	Same address	Low incremental cost to serve 2
Family	Households (up to 6 users)	\$16.99	Same address, user cap	Captures full household use
International	Users in lower-income	Varies (e.g., \$3, \$4)	ID/billing region	Adjusted to local

Case Study: Spotify's Direct Price Discrimination Strategy

Key Price Discrimination Tactics:

- **Student Plan:**
Validates with .edu and ID system (SheerID) → direct pricing based on age/income.
- **Family Plan:**
Ties users to same address → prevents resale/arbitrage of cheaper access.
- **Geographic Pricing:**
Varies pricing country by country → based on average income and digital spending.

Strategic Benefits

- Expands total user base by offering affordable entry points
 - Increases retention by offering upsell paths (e.g., Student → Individual → Duo)
 - Extracts maximum WTP (willingness to pay) from each group
 - Reduces churn by aligning price with perceived value
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- [Spotify Pricing](#), [Spotify Student Plan](#), [Spotify Premium Family Terms](#), [Spotify India Pricing](#)

Discussion Activity 3

- What makes Spotify's pricing strategy an example of smart discrimination? Could it backfire?
- Consider family plans, geography, and product usage. Is it fair? Transparent?

Activity – Student Discount Abuse

Scenario:

- A streaming service offers a 50% student discount on monthly subscriptions. To verify eligibility, students must upload a school email or student ID during signup. But the company has noticed:
 - Some users share login credentials with non-students
 - Others resell “verified” accounts
 - A few create fake email domains or doctored ID scans
- This undermines the price discrimination strategy, reduces revenue from high-WTP users, and may damage brand perception.

Your Task:

- As a pricing consultant, your team has been asked to redesign the student verification system to minimize abuse while keeping friction low for legitimate users.
- Design Considerations:
 - How can the firm ensure only students use the discounted plans?
 - How much verification is too much? What’s a fair tradeoff between security and convenience?
 - What technology or partnerships could be leveraged?
 - Could terms of use enforcement, device limits, or behavioral signals help?
- Propose a student verification strategy that reduces arbitrage risk while maintaining a good user experience. Explain why your approach balances cost, friction, and fraud prevention.

Activity – Student Discount Abuse

Possible strategies:

1. Partner with a Third-Party Verification Service

- Use platforms to validate enrollment directly with the student's school. These services:
 - Cross-reference names with enrollment databases
 - Offer APIs that reduce manual review
 - Are widely accepted in the industry

2. Device & Usage Limits

- Restrict account use to:
 - A maximum of 2 devices
 - No simultaneous streams (unless upgraded)
 - Monitor for unusual login patterns (e.g., multiple IPs or locations)

3. Annual Reverification

- Require users to reverify student status annually via the verification partner or .edu email expiration

4. TOS Enforcement & Messaging

- Clearly state in terms of use: "Accounts verified for student discounts must not be shared with non-students. Misuse may result in suspension or reversion to standard pricing."

Is Price Discrimination Illegal?

- It would seem that it is, but not necessarily for consumers.
- The Robinson-Patman Act (1936) closed the loophole of price discrimination from the Clayton Act (1914).
- The Robinson-Patman Act made it illegal for a business to sell or receive a good at a price discount from another business.
- In essence, a firm receiving any cut in price or giving any cut in price to another firm is illegal
- Large retailers (e.g. Walmart) and their suppliers are often accused of this.
- Combatting the issue:
 - Say it was a cost issue
 - Claim it was done in the face of competition
 - Allow promotional allowances
- How do economists feel about it?
 - It protects competitors rather than competition

Price Discrimination and the Robinson-Patman Act

Action	Is It Legal?	Explanation
Offering different prices to different buyers due to cost savings (e.g., bulk shipping, scale)	Legal	Permitted if cost justification can be documented
Offering lower prices to meet a competitor's offer	Legal	Allowed under "meeting competition" defense
Offering different prices based on customer group (e.g., student, senior)	Legal	Legal in consumer markets where resale isn't involved
Offering different prices to buyers of identical goods, with no cost or competitive justification	Conditionally Legal	Risky if the effect is to substantially harm competition
Favoring large retailers with better pricing, rebates, or promotional support not offered to smaller competitors	Illegal	Classic violation — discriminatory pricing must not harm competition between buyers
Discriminating in promotional allowances (e.g., ad money, display support) between resellers	Illegal	Must offer proportionally equal support to all competing customers

Key Takeaways

- Direct price discrimination charges different prices to consumer groups based on observable traits (e.g., age, location).
- Profit increases when firms capture more of each group's willingness to pay.
- Common implementations:
 - Student and senior discounts
 - Geographic pricing
 - Business vs. personal plans
 - In-store vs. online pricing
- Prevention of arbitrage is critical, use digital tools, ID verification, address requirements.
- Not all price differences are discriminatory, cost-based pricing is allowed.
- Robinson-Patman Act prohibits unjustified business-to-business discrimination in pricing goods (not services).
- Real-world pricing strategies must balance fairness, profitability, and legal risk.

Knowledge Check

1. What are the three conditions needed for direct price discrimination to succeed?
2. Why do businesses offer senior or student discounts?
3. What is arbitrage, and how does it threaten a pricing strategy?
4. How does the Robinson-Patman Act restrict business-to-business price discrimination?
5. What's the main benefit of price discrimination for firms?

Knowledge Check Answers

1. Market power, demand elasticity differences, and arbitrage prevention
2. To segment by elasticity (sensitive to price) and extract more consumer surplus
3. Arbitrage = buying low and reselling high; it undermines the pricing structure
4. It prohibits unjustified discounts between businesses unless cost-based or competitive
5. It increases profit by capturing more of the consumer's willingness to pay

Consultant's Challenge: Pricing for Profit

Scenario:

- You're consulting for a SaaS (software-as-a-service) company that wants to increase revenue by introducing direct price discrimination — including a student discount, nonprofit pricing, and international versions of their platform.

Your Task:

1. How would you help them design a price discrimination strategy that increases revenue while avoiding arbitrage?
2. What risks would you help them anticipate around customer fairness, legal limits (e.g., B2B pricing), or brand image?